# FOUR EQUIVALENT CHARACTERIZATIONS OF RIG-ETALE MORPHISMS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the proof is the standard property-defines-property-morphisms template with locality of rig-etale-ness as the input property

*Editorial note.* The label lemma-rig-etale-morphisms names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- the definition of rig-etale morphisms of locally Noetherian formal algebraic spaces via etale covers and rig-etale arrows of WAdm^Noeth (definition-rig-etale)
- rig-etale-ness is a local property of arrows of WAdm^Noeth (lemma-rig-etale-axioms of the same chapter)
- the Formal Spaces lemma that a local property of arrows defines a property of morphisms (formal-spaces-lemma-property-defines-property-morphisms) in its Noetherian variant (formal-spaces-remark-variant-Noetherian)

*Source reference.* Original reference 0AQM: sources/stacks-restricted/pinned-original-source.tex, source lines 5027–5027. Prerequisite pointer invoked at proof line(s) 5083; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original definition 0AIM: sources/stacks-chow/tags.txt, source lines 13270–13270. Prerequisite pointer invoked at proof line(s) 5065; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0ANG: sources/stacks-chow/tags.txt, source lines 13439–13439. Prerequisite pointer invoked at proof line(s) 5089; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original remark 0ANI: sources/stacks-chow/tags.txt, source lines 13441–13441. Prerequisite pointer invoked at proof line(s) 5092; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

---

*Date*: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original reference 0AQL: sources/stacks-restricted/pinned-original-source.tex, source lines 4788–4788. Prerequisite pointer invoked at proof line(s) 5086; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $S$ be a scheme. Let $f : X \to Y$ be a morphism of locally Noetherian formal algebraic spaces over $S$. The following are equivalent*

(1) *$f$ is rig-étale,*
(2) *for every commutative diagram*

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

*with $U$ and $V$ affine formal algebraic spaces, $U \to X$ and $V \to Y$ representable by algebraic spaces and étale, the morphism $U \to V$ corresponds to a rig-étale map in $WAdm^{Noeth}$,*
(3) *there exists a covering $\{Y_j \to Y\}$ as in Formal Spaces, Definition 0AIM and for each $j$ a covering $\{X_{ji} \to Y_j \times_Y X\}$ as in Formal Spaces, Definition 0AIM such that each $X_{ji} \to Y_j$ corresponds to a rig-étale map in $WAdm^{Noeth}$, and*
(4) *there exist a covering $\{X_i \to X\}$ as in Formal Spaces, Definition 0AIM and for each $i$ a factorization $X_i \to Y_i \to Y$ where $Y_i$ is an affine formal algebraic space, $Y_i \to Y$ is representable by algebraic spaces and étale, and $X_i \to Y_i$ corresponds to a rig-étale map in $WAdm^{Noeth}$.*

*Proof.* The equivalence of (1) and (2) is Definition 0AQM. The equivalence of (2), (3), and (4) follows from the fact that being rig-étale is a local property of arrows of $\mathrm{WAdm}^{Noeth}$ by Lemma 0AQL and an application of the variant of Formal Spaces, Lemma 0ANG for morphisms between locally Noetherian algebraic spaces mentioned in Formal Spaces, Remark 0ANI. □